

assignment kinematics1D - part 2

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:27pm

Quiz Instructions

2 attempts . print out the equation sheet first (first module in Canvas)

There is a tutorial in my dropbox but try hard first.



Question 1 10 pts

A car accelerates from 5m/s to 21m/s at a rate of 3m/s/s How far does it travel while accelerating?



117 m



207 m



69 m



41 m



Question 2 10 pts

A car is moving with a speed of 32.0 m/s. The driver sees an accident ahead and slams on the brakes, causing the car to slow down with a uniform acceleration of magnitude 3.50 m/s^2 . How far does the car travel after the driver put on the brakes until it comes to a stop?



4.57 m



9.14 m



292 m



112 m



146 m



Question 3 10 pts

A car increases its forward velocity uniformly from 40 m/s to 80 m/s while traveling a distance of 200 m. What is its acceleration during this time?

☐12 m/s²☐24 m/s²☐8.0 m/s²☐9.6 m/s²☐

Question 4 10 pts

An object is moving in a straight line with constant acceleration. Initially it is traveling at 16 m/s. Three seconds later it is traveling at 10 m/s. How far does it move during this time?

☐

30 m

☐

57 m

☐

39 m

☐

48 m

☐

Question 5 10 pts

An airplane needs to reach a velocity of 203.0 km/h to take off. On a 2000 m runway, what is the minimum acceleration necessary for the plane to take flight?

☐0.95 m/s²☐1.0 m/s²☐0.79 m/s²☐0.87 m/s²☐

Question 6 10 pts

A car is moving with a constant acceleration. At time $t = 5.0$ s its velocity is 8.0 m/s in the forward direction, and at time $t = 8.0$ s its velocity is 12.0 m/s forward. What is the distance traveled in that interval of time?

☐

20 m

☐

30 m

☐

50 m

☐

10 m

☐

40 m



Question 7 5 pts

A ball rolls across a floor with an acceleration of 0.100 m/s^2 in a direction opposite to its velocity. The ball has a velocity of 4.00 m/s after rolling a distance 6.00 m across the floor. What was the initial speed of the ball?

☐

5.85 m/s

☐

3.85 m/s

☐

5.21 m/s

☐

4.15 m/s

☐

4.60 m/s



Question 8 5 pts

Assuming equal rates of acceleration in both cases, how much further would you travel if braking from 56 mi/h to rest than from 28 mi/h

☐

4.8 times farther

☐

4 times farther

☐

5.2 times farther

☐

3.2 times farther



Question 9 5 pts

A speeding car is traveling at a constant 30.0 m/s when it passes a stationary police car. If the police car delays for 1.00 s before starting, what must be the magnitude of the constant acceleration of the police car to catch the speeding car after the police car travels a distance of 300 m?

7.41 m/s²1.45 m/s²6.00 m/s²3.70 m/s²3.00 m/s²

Question 10 5 pts

Acceleration is sometimes expressed in multiples of g , where $g=9.8\text{m/s/s}$ is the acceleration due to the earth's gravity. In a car crash, the car's velocity may go from 30m/s to 0m/s in 0.15s How many g 's are experienced, on average, by the driver?

20 g 26 g 24 g 14 g 

Question 11 5 pts

A train moving in a straight line with an initial velocity of 0.50 m/s accelerates at 2.0 m/s^2 for 2.0 s , coasts with zero acceleration for 3.0 s , and then accelerates at -1.5 m/s^2 for 1.0 s .

What is the final velocity?

hint: divide the motion in 3 phases. It will help to make a graph. For extra help see the slides. When $a=0$, the velocity does not change. dah

☐

4m/s

☐

9m/s

☐

3m/s

☐

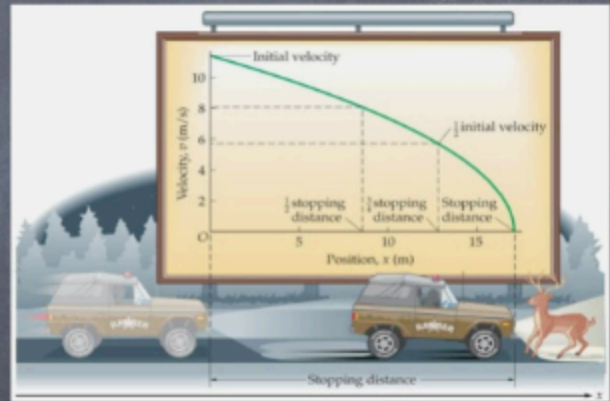
6m/s

☐

Question 12 5 pts

A park ranger driving at 11.4 m/s in back country suddenly sees a deer "frozen" in the headlights. He applies the brakes and slows with an acceleration of -3.80 m/s^2 .

- (a) If the deer is 20.0 m from the ranger's car when the brakes are applied, how close does the ranger come to hitting the deer?
- (b) What is the stopping time?



hint; the final velocity is 0

hint; the final velocity is 0

- ☐ 5m, 4s
- ☐ 2.9m, 2s
- ☐ 2.9m, 3s
- ☐ 17m, 3s



Question 13 5 pts

average speed= TOTAL distance / total time

You walk the first 0.20 km at 1.0 m/s and the next 0.20 km at 2.0 m/s. What is your average speed? (Be careful.)

Your average speed is _____ m/s.



- ☐ 7.3
- ☐ 0
- ☐ 1.3
- ☐ 3.1
- ☐ 5.2
- ☐

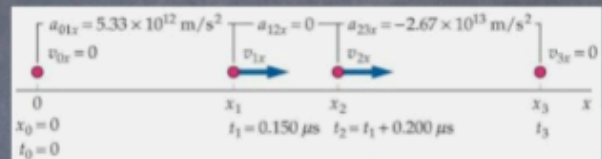
Question 14 5 pts

An electron in a cathode-ray tube accelerates from rest with a constant acceleration of $5.33 \times 10^{12} \text{ m/s}^2$ for $0.150 \mu\text{s}$, then drifts with a constant velocity for $0.200 \mu\text{s}$, then slows to a stop with a negative acceleration of $-2.67 \times 10^{13} \text{ m/s}^2$. (Note: $1 \mu\text{s} = 10^{-6} \text{ s}$).

How far does the electron travel?



Draw the electron (as a dot).



What is the total displacement of the electrons (add the displacements of phase1, phase2, phase3). For phase1, phase3 the initial velocity changes. Except phase 2.

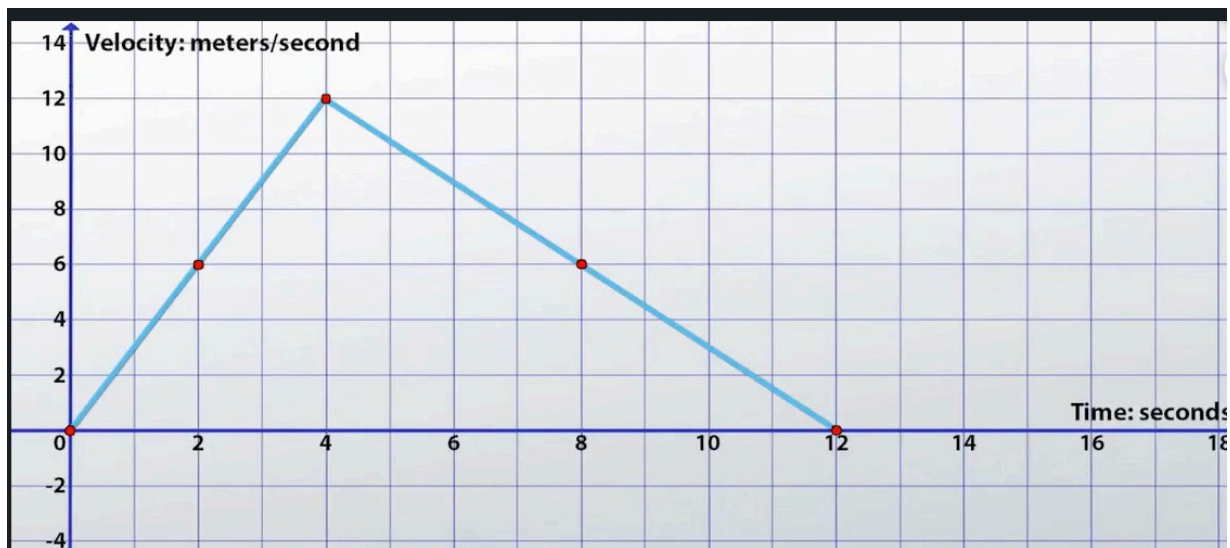
What is the total displacement of the electrons (add the displacements of phase1, phase2, phase3). For phase1, phase3 the initial velocity changes. Except phase 2.

- ☐ about 50cm
- ☐ about 30cm
- ☐ about 20cm
- ☐ about 100cm
- ☐



Question 15 5 pts

bonus. yey



What is the total displacement of the car (see calc minute in side). To find displacement from $V(t)$...

☐

12m

☐

72m

☐

48

☐

36m

☒

Question 16 2 pts

Bonus

What is the acceleration between 4s and 12s

☐

3m/s/s

☐

-3m/s/s

☐

-1.5m/s/s

☐

1.5m/s/s

Not saved

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